Answer Key

1. The speed of the centre of mass of the moving part is only $\frac{3}{4}$!

   $F = \frac{1}{2}$

2. $\ell_0 = 13$ m.

   Maximum speed of the jumper is $v = 18\ \text{ms}^{-1}$

   Maximum acceleration is $4g$.

3. The sphere cannot remain in equilibrium on the 30° inclined plane.

4. The period of the binary star therefore equals that of the Earth around the Sun divided by $\sqrt{2}$, *i.e.* $8\frac{1}{2}$ months.

5. *A* clearly failed his examinations.

6. $h_0 = \frac{2R^3}{3r^2} + \frac{2R^2 + r^2}{3r^2}\sqrt{R^2 - r^2} - \frac{m}{r^2\pi\rho}$.

   Do not forget that the formula above is only valid if the top of the ball is still covered by water

7. $v_1^{\min} = \sqrt{2gR\left(1+\sqrt{2}\right)} \approx 2.2\ \text{ms}^{-1}$.

8. $\mu = \sqrt{2} - 1 \approx 0.414$.

9. $\mu = \sqrt{Mm}$.

   The height attained by the final ball is $\frac{16M^2h}{\left(\sqrt{M}+\sqrt{m}\right)^4}$.

10. In a state of equilibrium, the water level does not change. The condition for this is

    $x = x_e = \left(\frac{T_2}{2T_1}\right)^2$.

    It can also be seen that overflow can only be a danger if it takes more than twice as long to empty the basin, as to fill it ($T_2 > 2T_1$).

11. $n_g^2 - n_w^2 \geq \cos^2\theta(n_g^2 + 1 - 2n_w)$.

    Substitution of the given values for the refractive indices yields the stated result

12. Thus, the period of a pivoted rod of length ℓ is approximately 82 per cent of that of a simple pendulum of the same length. This conclusion is valid not only for horizontal release, but for any common initial starting position.

13. $F = \frac{\pi D^2}{4}\frac{2\gamma}{d}$

14. The velocity of the thread at a distance of x metres from the wall is obviously proportionately smaller than the velocity of the end of the thread, *i.e.* it is xv_0.

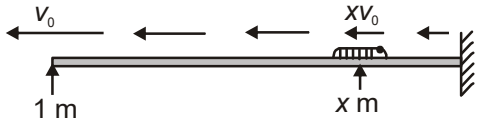


Fig. 23

If this value is greater than the speed of the caterpillar, then the latter will move away from the wall. Its situation will become more and more hopeless, and it will never reach the wall.

On the other hand, if $v_{caterpillar} > xv_0$, the net velocity of the caterpillar is towards the wall and increases as time passes, with the consequence that the caterpillar will certainly reach the wall. The limiting case corresponds to $x = v_{caterpillar}/v_0 = 0.1$ m. Starting at this point, the caterpillar does not move in either direction.

15. $\sin\theta = \frac{1}{\sqrt{3}}$,

16. When the amplitude is doubled (2 cm), the period is halved (1 s).

17. The body is under the point of suspension when $x(t) = 0$ and $y = y_0 = \frac{mg}{k}$,

18. The bodies on the wedge (the glass and the water in it) 'feel' as if they were in a gravitational field perpendicular to the inclined plane, with the consequence that the surface of the water lies parallel to the plane.

19. $M_n = \frac{m}{4}\sum_{k=1}^{n-1}\frac{1}{\sin(\pi k/n)}$.

$T = \pi\sqrt{\frac{R^3}{8GM_n}}$

20. $v_{max} = \sqrt{\frac{2dMg\sin\alpha}{m}}$.

21. (i) $F = \left(9m + \frac{7}{2}M\right)\frac{\cos\phi}{1+\sin\phi}g \approx 79$ N.

(ii) $\Delta a = a_2 - a_1 = \frac{5}{2}\frac{\cos\phi}{1+\sin\phi}g$.

(iii) $t = \sqrt{\frac{L}{\Delta a}} = 0.55$ s.

22. $\mu = \frac{1}{\pi}\ln 2 \approx 0.22.$

23. $t_1 + t_2 = \frac{4}{3}t_1 = 1.04\ \text{s}$

24. 'loop-wave' moves with speed $c = \sqrt{\frac{F}{\rho}}$, identical to the speed at which small transverse waves would propagate along the same rope.

The quantity $K = 2\pi Fc$ is a constant, characteristic of the rope and the tension in it. Of course, several (n) loops can be simultaneously excited and their total energy is

$$E_n(\omega) = nK\frac{1}{\omega} \quad (n = 0,1,2,3,...).$$

The momentum of the loop(s) can be calculated in a similar way,

$$P_n(\omega) = n(2\pi R\rho)c = nK\frac{1}{c\omega} \quad (n = 0,1,2,3,...),$$

as can the angular momentum

$$J_n(\omega) = \pm n(2\pi R\rho)Rc = \pm nK\frac{1}{\omega^2} \quad (n = 0, 1, 2, 3, ...).$$

25. The work done by the engine- taking only the acceleration of the sand into account – is 50 J s^{-1}, *i.e.* its power output is 50 W.

This means that one-half of the power of the engine (25 W) is converted into kinetic energy of the sand, the rest is the work done against friction and converted into heat.

26. $F = \frac{Q^2}{32\pi\varepsilon_0 R^4}\left(R^2 - h^2\right).$

27. **To Prove**

28. $Q_1 = -Q\frac{d-x}{d}$ and $Q_2 = -Q\frac{x}{d}$

29. $n(y) = \frac{R}{R-y}n_0.$

The material with the greatest known refractive index is diamond but even the refractive index of this material does not reach the value $n_{max} = 2.5$. It is this limit that sets the maximum angular size of the arc the light ray can cover. If the refractive index changes from $n_0 = 1$ to $n_{max} = 2.5$ then the maximum value of y is $\frac{3}{5}R$, corresponding to an arc of angular size 66.4°.

30. Compare the energies of the electrostatic fields of the dented and undented foils.

31. $R_e = \frac{2(c-1)}{nc}.$

32. $v_{max} = \frac{B\ell CV_0}{m + B^2\ell^2 C}$ and $Q_{min} = \frac{B^2\ell^2 C^2 V_0}{m + B^2\ell^2 C}$.

This means that the efficiency of the electromagnetic gun cannot be more than 25 per cent.

33. The lamp closest to the switch suddenly flashes, but the brightness of the two other lamps does not change. This all takes place in a very brief period and later all three lamps fade and go out.

34. $\rho = \frac{2\varepsilon_0 \omega m}{e}\left(\omega \pm \frac{eB}{m}\right)$

It is zero, if $\omega = \frac{|e|B}{m}$.

35. In case (*a*), $\lambda = \theta$ and solution of the simultaneous equations shows that *I*, and hence the voltmeter reading, is zero.

In case (*b*), $\lambda = \theta - \sin\theta$ and straightforaward but slightly lengthy algebra shows that the voltmeter reading will be

$$|V| = \frac{2\pi^2 Ra^2 \dot{B}\sin\theta}{4\pi^2 R + r\theta(2\pi - \theta)}.$$

36. $r = \sqrt{2}R$.

37. $\omega = -\frac{QB}{2m}$.

38. $v_0 = \frac{\mu_0 Q}{2m\pi} I \times \ln 2 \approx 2.46 \times 10^5\ \text{ms}^{-1} \approx 250\ \text{km s}^{-1}$.

Chemistry

Answer Key

1. (4)
2. (2)
3. (2)
4. (2)
5. (2)
6. (1)
7. (4)
8. (1)
9. (1)
10. (1)
11. (1 → P); (2 → R); (3 → S); (4 → Q)
12. Part I → 2; Part II → 2; Part III → 1
13. Part I → 4; Part II → 1; Part III → 4; Part IV → 3
14. (1 → P, Q, S); (2 → P, Q, R, S); (3 → P, Q, R, S); (4 → P, Q, R, S)
15. (3)
16. (4)
17. (3)
18. (4)
19. (5)
20. (4)
21. (4)
22. (1)
23. (1)
24. (2)
25. (2)
26. (3)
27. (1 → R); (2 → R); (3 → P); (4 → S)
28. (1 → R); (2 → P, Q); (3 → R, S); (4 → P, R, S)
29. (1 → Q); (2 → Q, S); (3 → P, Q, S); (4 → Q, S)
30. (A → 4); (B → 1)
31. (1) – 16; (2) – 4; (3) – 16; (4) – 4; (5) – 4; (6) – 4; (7) – 4; (8) – 3; (9) – 4; (10) – 4; (11) – 2
32. (2)
33. (2)
34. (2)
35. (3)
36. (2)
37. (2)
38. (4)
39. (2)
40. (4)
41. (3)
42. (2)
43. (1)
44. (3)
45. (1 → P); (2 → S); (3 → Q); (4 → R)
46. (1)
47. (2)
48. (2)
49. (2)
50. (3)
51. (2)
52. (2)
53. (1)

54 (2)

55. (3)

56. (2)

57. (1)

58. (4)

59. (2)

60. (3)

61. (2)

62. (2)

63. (3)

64. (3)

65. (2)

66. (4)

67. Reaction 1: B, D; Reaction 2: E, F, C
Reaction 3: I, A
Reaction 4: L, G; Reaction 5: B, L, C

68. (1 → R, S → Z); (2 → P, S → Y); (4 → Q, S → Y)

69. (1 → Q); (2 → P); (3 → S), (4 → R)

70. (2)

71. (3)

72. (2)

73. (2)

74. (2)

75. (3)

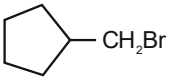
76. (4)

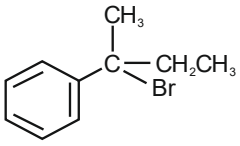
77. (4)

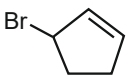
78. (2)

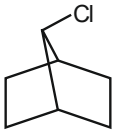
79. (2)

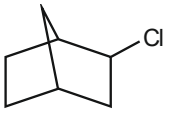
80. (3)

81. (1) 

(2) 

(3) 

(4) 

(5) 

Basic information.

82. (1) (A) Reaction faster due to 3° R – I

(2) (D) More reactive due to S_N2 reaction

(3) (E) More reactive due to S_N2 reaction

(4) (G) More reactive due to good leaving group

(5) (I) More reactive due to resonance stablized transition state formed.

83. (1 → q ; 2 → p ; 3 → r ; 4 → q)

84. (4)

85. (2)

86. (3)

87. (3)

88. (2)

89. (1 → P, Q, S); (2 → P, R); (3 → Q, S); (4 → R)

90. (1 → P); (2 → Q); (3 → R); (4 → S)

91. Substitution-(4), (6); Elimination-(1), (2), (3), (5)

3° alkyl halide, do not show S_N2

92. (3)

93. (4)

94. (3)

95. (2)

96. (2)

97. x = 4

98. (1)

99. (1)

100. (1)

101. (1)

102. (3)

103. (2)

104. (1)

105. (2)

106. (4)

107. (2)

108. (3)

109. (2)

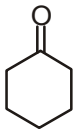
110. (3)

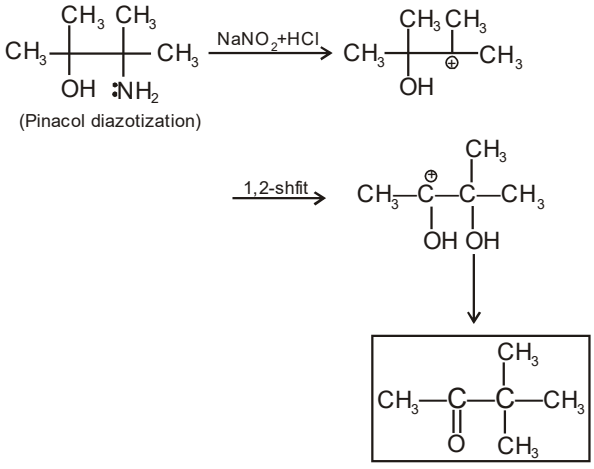
111. (1)

112. (1 → S); (2 → R); (3 → Q); (4 → P)

113. A – b; B – b; C – c; D – a

114. (2)

115.  (ring expansion)

116. 

117. (3)

118. (2)

119. (2)

120. (2)

121. (2)

122. (4)

123. (2)

124. (4)

125. (2)

126. (3)

127. (2)

128. (3)

129. (1)

130. (4)

131. (4)

132. (1 → P, R); (2 → R); (3 → S); (4 → P, R)

133. (1 → P, Q, S); (2 → P); (3 → P, Q, S); (4 → P, Q, S)

134. (1)

135. (1)

136. (2)

137. (2)

138. (1)

139. (2)

140. (2)

141. (1)

142. (4)

143. (1)

144. (1)

145. (2)

146. (2)

147. (3)

148. (1)

149. (2)

150. (2)

151. (1)

152. (3)

153. (1)

154. (2)

155. (2)

156. (2)

157. (3)

158. (1)

159. (3)

160. (3)

161. (2)

162. (4)

163. (2)

164. (3)

165. (2)
166. (4)
167. (1 → Q); (2 → P); (3 → Q); (4 → Q)
168. (2)
169. (1)
170. (2)
171. (1)
172. (2)
173. (3)
174. (2)
175. (2)
176. (4)
177. (4)
178. (2)
179. (1)
180. (3)
181. (4)
182. (3)
183. (3)
184. (4)
185. (3)
186. (2)
187. (3)
188. (2)
189. (2)
190. (3)
191. (3)
192. (4)
193. (2)
194. (3)
195. (4)
196. (3)
197. (1)

❑ ❑ ❑

MATHEMATICS

ANSWERS

1. (A) 2. (B) 3. (A) 4. (D) 5. $\frac{(-1)^n}{n+1}$

6. $\frac{2}{3}$ 7. $\frac{x}{3}+\frac{1}{3}$ 8. e^2 10. $\frac{\sqrt{3}}{2}$

11. $\frac{2L}{3}$ 13. e 14. $\frac{-\pi^2}{4}$ 15. 8 16. $\theta\cdot\cot\theta$

18. (B) 19. (B) 20 (A)

21. A → (P); B → (S); C → (Q); D → (R)

22. (D) 23. (D) 24. (B) 25. (C) 26. (A)

27. (A)

28. (A, B, C, D)

29. (B) 30. (C) 31. (C)

32. 1 33. 5 34. 1 35. 8

36. (D) 37. (A) 38. (C) 39. (B) 40. (C)

41. (A) 42. (C) 43. (A, C, D) 44. (A, C) 45. (A, D)

46. (B, C)

47. (A) 48. (C) 49. (B)

50. (A) 51. (C) 52. (B) 53. (C) 54. (B, C)

55. (A, B, C)

56. (C) 57. (B) 58. (D)

59. A → (P, Q); B → (P, Q, R, S); C → (P, S, T); D → (P, Q, R, S, T)

60. 4 61. 1 62. 9 63. 1 64. 9

65. 2

66. (B) 67. (C) 68. (A, D) 69. (C) 70. (A)

71. (B) 72. (B) 73. (D) 74. (C) 75. (A)

76. (B, C, D)

77. A → (S); B → (Q); C → (P); D → (Q)

78. (B) 79. (C) 80. (A) 81. (C) 82. (D)
83. (A) 84. (B) 85. (C) 86. (B) 87. (B)
88. (B) 89. (A) 90. (C) 91. (A, C) 92. (A, C)
93. (A, B) 94. (A, B, C, D) 95. (A, C)
96. (D) 97. (A) 98. (A)
99. A → (Q); B → (R); C → (P); D → (S) 100. A → (P); B → (Q); C → (P); D → (S)
101. A → (Q); B → (S); C → (P); D → (R)
102. 4 103. 8
104. (D) 105. (C) 106. (D) 107. (B) 108. (D)
109. (D) 110. (B) 111. (C) 112. (C) 113. (B)
114. (D) 115. (C) 116. (D) 117. (A) 118. (B, C)
119. (A, B) 120. (A, B, C) 121. (A, B, C, D) 122. (A, C)
123. (B) 124. (B)
125. (C) 126. (B) 127. (A)
128. A → (Q); B → (P); C → (S); D → (R)
129. A → (Q); B → (S); C → (P); D → (R)
130. 3 131. 21 132. 6 133. 7 134. 3

❑ ❑ ❑